

Experimental Methods

Cheatsheet for IBMS3

DNA

PCR

- **Principle:** Exponential amplification of specific DNA fragments via thermal cycling (denaturation, annealing, extension) with DNA polymerase.
- **Applications:** Gene cloning, pathogen detection, genetic disease diagnosis.
- **Data:** Gel band pattern by fragment size; summarized as band present/absent and band intensity at the expected bp.

qPCR (*Quantitative PCR — detect quantity & expression level*)

- **Principle:** Real-time monitoring of fluorescent signal intensity to quantify DNA amplification.
- **Categories:** SYBR Green detection (a DNA dye that binds the dsDNA proportionally) and TaqMan-probe (exonuclease activity of Taq polymerase releases the reporter from the quencher).
- **Applications:** Gene expression analysis, pathogen load quantification.
- **Data:** Amplification curve (fluorescence vs cycle); summarized by Ct/Cq, $\Delta\Delta Ct$, fold change, or copy number.

Southern Blotting (*provides structural information*)

- **Principle:** DNA separated by gel electrophoresis, transferred to membrane, hybridized with labeled probes to detect target sequences.
- **Applications:** Gene copy-number analysis, RFLP detection.
- **Data:** Membrane band pattern for DNA fragments; summarized by band size, band intensity, or copy-number estimate.

RNA

RT-PCR (*Reverse Transcription PCR*)

- **Principle:** RNA reverse-transcribed to cDNA, then amplified by PCR.
- **Applications:** Low-abundance RNA detection, viral RNA quantification.
- **Data:** Usually gel band pattern after cDNA amplification; if run as RT-qPCR, then a qPCR-style amplification curve with Ct/Cq.

Northern Blotting

- **Principle:** RNA separated by gel electrophoresis, transferred to membrane, hybridized with labeled probes to detect target transcripts.
- **Applications:** Gene expression analysis, RNA splicing studies.
- **Data:** Membrane band pattern for RNA transcripts; summarized by transcript size and relative band intensity.

FISH

- **Principle:** A probe-based method for RNA visualization.
- **Applications:** Cellular-spatial localization.
- **Data:** Fluorescence microscopy images with puncta/signals; summarized as puncta per cell, percent positive cells, or spatial localization.

Protein

Western Blotting

- **Principle:** Proteins separated by gel electrophoresis, trans-

ferred to membrane, detected with specific antibodies; qualitative and semi-quantitative.

- **Applications:** Protein expression quantification, post-translational modification (PTM) analysis.
- **Data:** Protein band pattern on blot; summarized by band size plus normalized densitometry.

Co-IP (*Co-Immunoprecipitation — in vivo detection*)

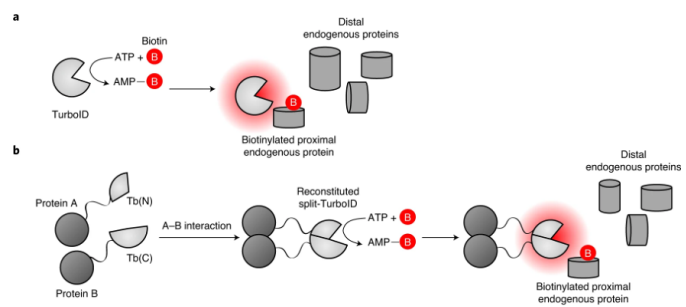
- **Principle:** Antibody precipitates target protein together with interacting partners; complex identified by mass spectrometry (MS).
- **Applications:** Protein-protein interaction network studies.
- **Data:** Western blot bands or mass-spec interactor list after pull-down; summarized as interactor presence or enrichment over control.

Pull-down Assay (*in vitro; e.g. novel drug interactions*)

- **Principle:** Recombinant tagged bait protein (e.g. GST) captures interacting proteins; identified by MS. Adaptable to RNA pull-down or DNA pull-down formats.
- **Applications:** Validating known or screening novel protein interactions.
- **Data:** Bound-protein bands or mass-spec interactor list; summarized as specific binding or enrichment over control.

Proximity labeling with TurboID

- **Principle:** TurboID uses ATP and biotin to generate biotin-5'-AMP, a reactive intermediate that can covalently label proximal endogenous proteins to study the spatial localization and interactions of proteins in the cell.
- **Data:** Biotinylated protein enrichment list from MS; summarized as enriched proximal proteins and fold enrichment.



Mass Spectrometry (MS)

- **Principle:** Proteins ionized and separated by mass-to-charge ratio; identified by database matching.
- **Applications:** Proteomics, PTM analysis.
- **Data:** m/z spectra leading to peptide/protein abundance tables; summarized as peptide IDs, protein abundance, or PTM site intensity.

FRET (*Fluorescence Resonance Energy Transfer*)

- **Principle:** Energy transfer occurs when two fluorescently labeled proteins are in close proximity, reporting on interaction.
- **Applications:** Dynamic protein-protein interaction monitoring in living cells.
- **Data:** Donor/acceptor fluorescence ratio trace or image map; summarized by FRET efficiency or interaction dynamics over time.

ELISA (*Enzyme-Linked Immunosorbent Assay*)

- **Principle:** Enzyme-linked immunoassay; uses colorimetric detection for quantitative analysis of proteins in solution (e.g. antibodies).
- **Applications:** Quantitative protein detection in serum, cell

media, etc.

- **Data:** Standard curve + absorbance/OD values; summarized as concentration such as pg/mL or ng/mL.

Protein–DNA Interaction Methods

ChIP-seq (*Chromatin Immunoprecipitation Sequencing*)

- **Principle:** Crosslink protein–DNA complexes, immunoprecipitate, then sequence to map binding sites genome-wide.
- **Applications:** Transcription factor binding mapping, epigenetic regulation studies.
- **Data:** Genome-aligned reads / peak tracks; summarized as peaks, read density, or occupancy at loci.

Footprinting Assay

- **Principle:** Protein-bound regions protect DNA from enzymatic or chemical cleavage; protected sites revealed by sequencing.
- **Applications:** Precise mapping of protein–DNA interaction interfaces.
- **Data:** Cleavage protection profile along DNA; summarized as protected bases or footprinted region.

EMSA (*Electrophoretic Mobility Shift Assay*)

- **Principle:** Protein–DNA complexes migrate more slowly in gel electrophoresis than free DNA.
- **Applications:** Qualitative detection of protein–DNA interactions.
- **Data:** Shifted vs free band pattern on gel; summarized as bound/unbound fraction or shift intensity.

CUT&RUN (*Cleavage Under Targets and Release Using Nuclease*)

- **Principle:** Antibody-targeted micrococcal nuclease cleaves protein-bound DNA fragments; enables high-resolution mapping from low input material.
- **Applications:** Chromatin accessibility analysis, transcription factor binding mapping.
- **Data:** Genome-aligned reads / peak tracks; summarized as high-resolution binding peaks or occupancy signal.

Protein–mRNA / Translatome Detection

Ribosome Profiling (*Ribo-seq*)

- **Principle:** Capture ribosome-associated mRNA, degrade ribosomes, then sequence the protected mRNA fragments. Provides a snapshot of active translation, including which mRNAs are being translated, ribosome position on the transcript, and translation efficiency per gene.
- **Applications:** Monitoring mRNA–protein binding during translation.
- **Data:** Ribosome-footprint read matrix / codon occupancy tracks; summarized as translation efficiency or ribosome density.

Gene Expression Sequencing

RNA-seq (*RNA Sequencing*)

- **Principle:** RNA reverse-transcribed to cDNA and sequenced at high throughput to quantify gene expression.
- **Applications:** Differential expression analysis, novel transcript discovery.
- **Data:** Gene/transcript count matrix; summarized as counts, TPM/FPKM, or log2 fold change.

Single-cell RNA-seq (*scRNA-seq*)

- **Principle:** Gene expression profiling at single-cell resolution,

revealing cellular heterogeneity.

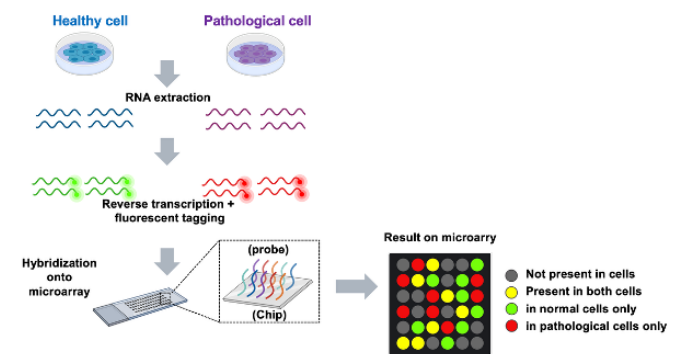
- **Applications:** Tumor microenvironment, stem cell differentiation studies.
- **Data:** Cell × gene count matrix; summarized as clusters, marker genes, and per-cell expression.

Spatial Transcriptomics

- **Principle:** Gene expression profiling that preserves spatial information within tissue sections.
- **Applications:** Tissue development, tumor spatial heterogeneity studies.
- **Data:** Spot/cell × gene count matrix plus spatial coordinates; summarized as spatial expression maps.

Microarray

- **Principle:** Hybridization of target sequences to specific DNA probes on a chip to measure expression levels.
- **Applications:** Multi-gene expression profiling, clinical diagnostics.
- **Data:** Probe intensity matrix; summarized as normalized probe intensity and differential expression.



Long-read Sequencing (*3rd-generation sequencing*)

- **Principle:** Sequences full-length transcripts, retaining splice variant and structural information; also detects structural variants (repeat expansions, translocations, inversions).
- **Applications:** Complex transcript structure, novel splice variant discovery, genomic SV detection.
- **Data:** Full-length read alignments / isoform table; summarized as isoform counts, splice variants, or structural variant calls.

Pro-seq (*Precision Run-On Sequencing*)

- **Principle:** Improvement over conventional RNA-seq; enriches and sequences nascent RNA being actively transcribed by RNA polymerase.
- **Applications:** Real-time transcriptional activity profiling.
- **Data:** Nascent-transcription read density tracks; summarized as polymerase occupancy or transcription-rate proxy.

Epigenetic Sequencing

Bisulfite Sequencing

- **Principle:** Bisulfite converts unmethylated cytosines to uracil; methylation patterns resolved by comparison with reference after sequencing.
- **Applications:** Genome-wide DNA methylation mapping.
- **Data:** Per-CpG methylation calls; summarized as percent methylation at CpGs or regions.

ATAC-seq (*Assay for Transposase-Accessible Chromatin with Sequencing*)

- **Principle:** Tn5 transposase inserts into open chromatin re-

gions and adds sequencing adapters.

- **Applications:** Chromatin accessibility analysis, regulatory element localization.
- **Data:** Open-chromatin insertion / peak tracks; summarized as accessibility peaks or fragment counts.

m6A Sequencing

- **Principle:** Immunoprecipitation of N6-methyladenosine (m6A)-containing RNA fragments, then sequencing.
- **Applications:** Functional studies of RNA methylation modifications.
- **Data:** m6A-enriched peak tracks; summarized as modified regions and peak enrichment.

MeDIP-seq (*Methylated DNA Immunoprecipitation Sequencing*)

- **Principle:** Enriches methylated DNA using anti-5-methylcytosine antibodies before sequencing.
- **Applications:** Genome-wide methylation analysis, cancer epigenetics.
- **Data:** Methylated-region enrichment tracks; summarized as methylation peak enrichment.

MNase-seq (*Micrococcal Nuclease Sequencing*)

- **Principle:** Micrococcal nuclease digests linker DNA; nucleosomal DNA retained and sequenced.
- **Applications:** Nucleosome positioning and chromatin structure analysis.
- **Data:** Nucleosome occupancy / positioning tracks; summarized as nucleosome positions and occupancy scores.

DNase-seq (*DNase I Hypersensitive Sites Sequencing*)

- **Principle:** DNase I digests open chromatin; accessible sites identified by sequencing.
- **Applications:** Identification of regulatory elements and TF binding sites.
- **Data:** Cut-site / hypersensitive site tracks; summarized as DHS peaks or accessibility score.

FAIRE-seq (*Formaldehyde-Assisted Isolation of Regulatory Elements*)

- **Principle:** Formaldehyde crosslinking followed by phenol-chloroform extraction isolates nucleosome-free DNA regions for sequencing.
- **Applications:** Mapping active regulatory elements and open chromatin regions.
- **Data:** Open-chromatin enrichment tracks; summarized as accessible-region peak calls.

Small RNA-seq

- **Principle:** Isolation and sequencing of small non-coding RNA molecules.
- **Applications:** miRNA, siRNA, and piRNA expression profiling.
- **Data:** Small RNA count matrix; summarized as miRNA/siRNA/piRNA abundance.

mRNA Visualization & Analysis

FISH (*Fluorescence In Situ Hybridization*)

- **Principle:** Fluorescently labeled probes hybridize to RNA in cells/tissues; localization visualized by microscopy.
- **Applications:** RNA subcellular localization, viral RNA detection.
- **Data:** Fluorescence microscopy images with puncta/signals; summarized as puncta per cell, percent positive cells, or spatial localization.

CLIP (*Crosslinking and Immunoprecipitation*)

- **Principle:** UV crosslinking of protein–RNA complexes, immunoprecipitation, then sequencing.
- **Applications:** Nucleotide-resolution mapping of RNA-binding protein (RBP) targets.
- **Data:** RBP-binding peak tracks / crosslink-site map; summarized as binding sites and peak enrichment.

RNA Editing-based RNA-seq

- **Principle:** RBP–APOBEC fusion protein induces RNA editing; binding sites identified by sequencing-based differential analysis.
- **Applications:** RBP target identification.
- **Data:** Editing event table plus aligned reads; summarized as editing frequency per site.

Chromosome Structure Analysis

Hi-C

- **Principle:** Formaldehyde crosslinks chromatin; enzymatic digestion, ligation, and sequencing generates a genome-wide 3D interaction map.
- **Applications:** 3D genome organization, long-range regulatory element discovery.
- **Data:** Contact matrix of genomic bin $\times$ bin interactions; summarized as contact frequency, compartments, loops, or TADs.

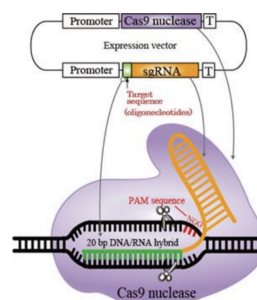
3C–5C (*Chromosome Conformation Capture variants*)

- **Principle:** A family of related techniques with differing resolutions for probing chromatin interactions (3C, 4C, 5C).
- **Applications:** Targeted or genome-wide chromatin interaction studies.
- **Data:** Targeted interaction-frequency data or matrix-like contacts; summarized as interaction strength between loci.

Gene Editing Technologies

CRISPR–Cas9 (*gene KO*)

- **Principle:** sgRNA guides Cas9 nuclease to cut target DNA. Cell repairs the break by NHEJ, causing bases to be removed or inserted, which leads to a codon shift and a NON-FUNCTIONAL PROTEIN. Inject vector containing guide RNA and mRNA for Cas9 into a one-cell mouse embryo
- **Applications:** Gene knockout, knock-in, base editing.
- **Data:** Edit genotype table; summarized as indel spectrum, editing rate, or knockout efficiency.



- **To get DSB in the right place**
- Design your **guide RNA** to be complementary to the region in your gene of interest that you want to mutate
- sgRNA includes a 'scaffold' that allows Cas9 to bind
- Region to be cut needs a nearby '**PAM sequence**' (5'-NGG-3'); used by Cas9 to destabilise DNA double helix so gRNA can bind
- Cas9 does not need to be designed, it is the same for every experiment: transcribed into protein

RNAi (*RNA Interference — Gene KD*)

- **Principle:** siRNA or shRNA targets mRNA for degradation, suppressing gene expression.
- **Applications:** Gene function studies, therapeutic silencing.
- **Data:** Knockdown readout from qPCR, western, or phenotype assay; summarized as percent knockdown and down-

stream phenotype.

CRISPRi (CRISPR Interference — gene repression)

- **Principle:** Catalytically dead dCas9 fused to a transcriptional repressor (e.g. KRAB) binds the target promoter and inhibits transcription without altering DNA sequence.
- **Applications:** Reversible gene silencing, essential gene studies, multi-gene regulation.
- **Data:** Expression repression readout; summarized as fold repression of the target gene.

CRISPRa (CRISPR Activation — gene activation)

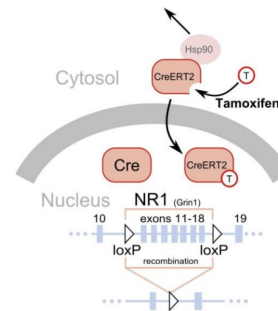
- **Principle:** dCas9 fused to transcriptional activator domains (e.g. VP64, p65) enhances transcription at target promoters.
- **Applications:** Endogenous gene activation, cell reprogramming, gain-of-function screens.
- **Data:** Expression activation readout; summarized as fold activation of the target gene.

CRISPR-based Epigenetic Editing (epigenome modification)

- **Principle:** dCas9 fused to epigenetic effector enzymes (e.g. DNMT3A, TET1, HDAC) to alter epigenetic state at specific genomic loci.
- **Applications:** DNA methylation editing, histone modification regulation, epigenetic disease research.
- **Data:** Epigenetic-state plus expression readout; summarized as chromatin/methylation change and expression shift.

Cre-loxP System (conditional gene knockout)

- **Principle:** Based on a recombinase system found in viruses in which an enzyme (Cre) cuts only at specific (loxP) sites. Cre recombinase recognizes loxP sites and catalyzes DNA recombination, enabling tissue-specific or temporally controlled gene deletion.
- **Applications:** Conditional knockout, cell lineage tracing, tissue-specific gene expression.
- **Data:** Recombination genotype readout; summarized as recombined vs unrecombined allele frequency.



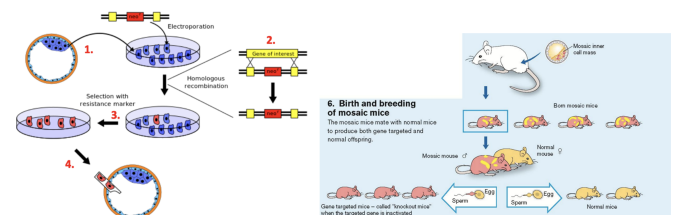
- Cre becomes activated in your cell type of interest in the embryo, but you would like to wait until adulthood.
- Use a Cre that has been modified to make it 'tamoxifen inducible' by fusion of a modified ER specific to tamoxifen (CreERT2)
- Cre is sequestered in cytoplasm until mice are given a tamoxifen injection, then it translocates to nucleus and knocks out gene

Tet-On/Tet-Off System (inducible gene expression)

- **Principle:** Tetracycline (or analog) modulates tTA/rtTA transcription factor activity to reversibly and dose-dependently control target gene expression.
- **Applications:** Inducible expression, time-specific functional studies, toxic gene control.
- **Data:** Inducible expression curve vs doxycycline or time; summarized as fold induction or repression.

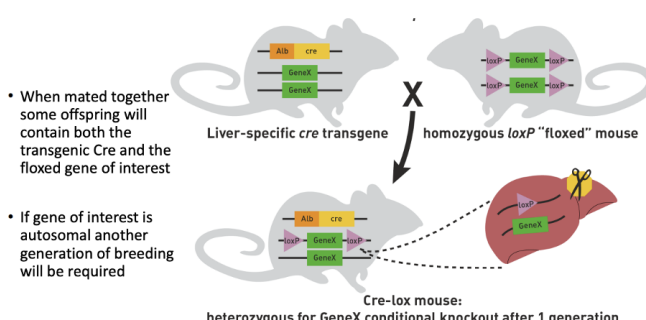
Homologous Recombination (HR) (precise gene modification)

- **Principle:** Exploits endogenous DNA repair machinery; homologous DNA templates enable precise gene modification.
- **Applications:** Gene knock-in, point mutation introduction, gene tag insertion.
- **Steps:** (i) Obtain wild-type embryonic stem cells from a blastocyst and grow them in culture; (ii) Electroporate cells to introduce a transgene that can KO gene of interest using homologous recombination (replace coding sequence with a neomycin resistance gene); (iii) Select cells with KO by treating with neomycin; (iv) Transfer knockout cells to the recipient blastocyst. Implant in the uterus to complete development; (v) Breeding of offspring to obtain heterozygous KO mice.
- **Data:** Correct-targeting genotype readout; summarized as knock-in or knockout confirmation rate.



- Two genetically modified mouse lines must be mated together:

- The **Cre line**
 - A transgene with Cre recombinase downstream by a cell-specific promoter
 - Usually randomly inserted into genome
- The **'floxed' line**
 - Has loxP sites around the gene that you want to knock out
 - Both copies need to be 'floxed'
 - (made by gene targeting)



- When mated together some offspring will contain both the transgenic Cre and the floxed gene of interest
- If gene of interest is autosomal another generation of breeding will be required

TALENs (Transcription Activator-Like Effector Nucleases — targeted DNA cleavage)

- **Principle:** TALE DNA-binding domains fused to FokI nuclease; used in pairs to create double-strand breaks.
- **Applications:** Gene knockout, knock-in, large genomic deletions.
- **Data:** Edit genotype table; summarized as editing efficiency and indel spectrum.

ZFNs (Zinc Finger Nucleases — first-generation programmable nucleases)

- **Principle:** Zinc finger protein DNA-binding domains fused to FokI nuclease; used in pairs to create double-strand breaks.
- **Applications:** Gene knockout, knock-in, cell therapy.
- **Data:** Edit genotype table; summarized as editing efficiency and indel spectrum.

CRISPR Screens

CRISPR Screen for Disease Genes

- *Goal:* Identify genes associated with disease onset or susceptibility.
- *Approach:* Genome-wide CRISPR–Cas9 knockout library screening.
- *Data:* sgRNA count matrix before/after selection; summarized as sgRNA enrichment/depletion and gene-level score.

CRISPR Screen for Drug Resistance Genes

- *Goal:* Identify genes associated with drug resistance.
- *Approach:* Genome-wide CRISPR–Cas9 knockout library screening.
- *Data:* sgRNA count matrix with and without drug; summarized as resistance-gene ranking and enrichment score.

Epigenome Engineering via CRISPR

- *Principle:* CRISPR–dCas9 fused to epigenetic modifier enzymes (e.g. DNMT3A) to introduce targeted epigenetic marks.
- *Applications:* Functional studies of DNA methylation and histone modifications.
- *Data:* Targeted epigenetic plus expression readout; summarized as locus-specific chromatin change and phenotypic effect.

Cellular & Physiological Methods

Flow Cytometry

- *Principle:* Laser-based technology detecting surface or intracellular protein distribution and levels in individual cells.
- *Applications:* Immune cell subset classification (immunophenotyping), cell sorting, phenotypic assessment of gene–disease relationships.
- *Data:* Per-cell event table with scatter and fluorescence channels; summarized as percent gated population, MFI, or cell counts.

FUCCI (*Fluorescence Ubiquitin Cell Cycle Indicator*)

- *Principle:* Two fluorescent proteins fused to cell-cycle regulators report cell cycle phase:
 - mVenus–hGem(1/110): yellow-green, expressed in S/G2/M phase
 - mCherry–hCdt1(30/120): red, expressed in G1 phase
 - Red+green simultaneously: transition phase
- *Applications:* Real-time cell cycle monitoring, proliferation assays, tumor and drug research.
- *Data:* Time-lapse fluorescence images or per-cell phase labels; summarized as fraction of cells in G1 vs S/G2/M.

EdU Detection (*S-phase labeling*)

- *Principle:* EdU (thymidine analog) incorporates into DNA during S-phase; detected by click chemistry without DNA denaturation. Combinable with FUCCI for precise S-phase identification.
- *Applications:* Cell proliferation assays, cell cycle studies.
- *Data:* Fluorescence images or flow events for EdU incorporation; summarized as percent EdU-positive cells or S-phase fraction.

ASO (*Antisense Oligonucleotides*)

- *Principle:* ASOs downregulate target gene expression at the RNA level.
- *Key feature:* Capable of crossing the blood–brain barrier.
- *Applications:* Investigating gene–phenotype relationships at the cellular level.
- *Data:* Knockdown plus phenotype readout; summarized as percent target reduction and phenotype change.

DREADD (*Designer Receptors Exclusively Activated by Designer Drugs*)

- *Principle:* Engineered G-protein coupled receptors that modulate neuronal activity in animal models in response to designer ligands with no endogenous targets.
- *Applications:* Chemogenetic control of neuronal circuits in vivo.
- *Data:* Physiology / behavior / activity traces; summarized as firing rate, calcium signal, or behavior score.

Genomics & Population Genetics

GWAS (*Genome-Wide Association Study*)

- *Principle:* Identifies SNPs associated with diseases or phenotypes across the genome; most significant hits reside in non-coding regions.
- *Applications:* Disease susceptibility loci discovery.

eQTL (*Expression Quantitative Trait Loci*)

- *Principle:* Links non-coding SNP loci (from GWAS) to gene expression levels, enabling functional interpretation of GWAS signals and prioritization of causal genes.
- *Applications:* Functional annotation of GWAS hits.

Luciferase Reporter Assay (*Dual-Luciferase System*)

- *Principle:* Tests whether a regulatory sequence (e.g. TE or enhancer) directly controls gene expression via a target promoter.
 - Validates TF–promoter regulatory relationships
 - Quantifies changes in transcriptional activity
 - Identifies key regulatory elements in promoter/enhancer regions
 - Probes signaling pathway effects on gene expression
 - Internal control (HSV-TK–*Renilla* luciferase) normalizes for transfection efficiency, cell number, and viability
- *Data:* Luminescence intensity or signal ratio; summarized as normalized reporter activity such as Firefly/*Renilla* ratio.

Animal & Model Organism Methods

Gene-Modified Mouse Models (*Cre/loxP & related systems*)

- *Principle:* Cre/loxP or other gene editing techniques generate mice with specific genetic modifications.
- *Applications:* Disease modeling, tissue-specific phenotypic characterization.
- *Data:* Genotype plus phenotype table; summarized as genotype frequencies and tissue/organism phenotype.

C. elegans Aging Model

- *Principle:* Mutant worm strains constructed; RNAi used to knock down target genes; survival analysis applied to explore aging mechanisms.
- *Applications:* Longevity gene identification, aging pathway research.
- *Data:* Lifespan / survival table; summarized as median lifespan or hazard ratio.

Microbiome Studies (*Host–Microbial Interactions*)

- *Principle:* Defined microbial communities introduced into mice; metabolic and phenotypic outcomes evaluated.
- *Applications:* Host–microbiome interaction, lipid metabolism research.
- *Data:* Microbial abundance table plus host phenotype data; summarized as relative abundance, metabolites, or phenotype.

Cold Thermogenesis Model (*Adipose Tissue & Tem-*

perature)

- *Principle:* Mice subjected to cold exposure; effects on adipose tissue examined by gene expression and protein analysis.
- *Applications:* Adipocyte biology, thermogenic gene regulation.
- *Data:* Expression/protein/physiology measurements; summarized as thermogenic gene expression, temperature, or body weight.

Structural & Computational Methods

Cryo-EM & Structural Biology

- *Principle:* High-resolution structural determination of macromolecular complexes (e.g. histone modification complexes) to understand the structural basis of epigenetic regulation and transcription.
- *Applications:* Basic epigenetic mechanism research, therapeutic target identification.
- *Data:* Particle images leading to a 3D density map / structure; summarized as resolution in Å and structural model quality.

Transformer-based Deep Learning (*Genomic modeling*)

- *Principle:* Deep learning architectures applied to genomic data for predicting mRNA translation, RNA-protein binding, somatic mutations in cancer, frameshift/non-synonymous/splice-site mutations, and tissue-specific splicing alterations from SNVs.
- *Applications:* Precision medicine, variant effect prediction.

Antibody De Novo Design (*Computational antigen design*)

- *Principle:* Algorithm-based antigen/antibody design pipeline integrated with wet-lab experimental validation.
- *Applications:* Immunotherapy development, novel therapeutic antibody generation.

Survival Analysis

- *Key methods:* Kaplan–Meier curves; Log-rank test.
- *Applications:* Clinical outcome analysis, animal model longevity studies.

Case Studies & Integrated Pipelines

Hemoglobin Switching (BCL11A): Gene editing → Functional validation → Epigenetic mechanism elucidation. Study of BCL11A's role in regulating the switch from fetal hemoglobin (HbF) to adult hemoglobin (HbA). Methods: CRISPR editing, ChIP-seq, luciferase reporter assay, Pro-seq.

Lymphoma Immunotherapy Development Pipeline:

1. Functional genomics target screening (CRISPR screens)
2. Preclinical model establishment
3. CAR-T cell development
4. Immune checkpoint inhibitor research
5. Clinical trial design and analysis